

Foundations of Artificial Intelligence

Part VII: Introduction to Robotics & Computer Vision (Overview)

Department of Computer Science

Lecture Roadmap

1. Robotics and Vision Overview
2. Perception
3. Sensors
4. Image Classification (Basic Idea)
5. AI in Autonomous Systems

Learning Objectives

By the end of this lecture, you should be able to:

- Explain how robotics combines perception, reasoning, and action.
- Identify common sensors used by intelligent robots and autonomous systems.
- Describe the basic idea of image classification in computer vision.
- Discuss how AI enables autonomy in self-driving and other robotic platforms.
- Recognize practical limitations: uncertainty, safety, and real-time constraints.

Robotics and Vision Overview

What Is Robotics in AI?

Robotics: Building agents that sense, reason, and act in the physical world.

Core components:

- **Perception:** Understand environment from sensors.
- **Decision-making:** Plan and choose actions.
- **Control:** Execute actions through motors/actuators.

Computer Vision is a key part of robotic perception:

- Detect lanes, objects, humans, obstacles.
- Estimate scene structure and motion.
- Support safe navigation and interaction.

Why Robotics + Computer Vision Matters

Real systems where AI and CV are essential:

- Autonomous driving and driver-assistance.
- Warehouse and logistics robots.
- Surgical and assistive medical robots.
- Drones for mapping, inspection, and delivery.
- Smart manufacturing and quality control.

High-level loop:

Sense → Understand → Plan → Act → Repeat

A Canonical Robotics Pipeline

Stage	Typical Tasks
Sensing	Capture data using camera, LiDAR, radar, IMU, GPS, ultrasound.
Perception	Object detection, localization, scene understanding, tracking.
Mapping	Build map or occupancy grid from observations.
Planning	Path planning and behavior selection under constraints.
Control	Convert plans into steering, velocity, and actuator commands.
Feedback	Re-evaluate after each action to handle uncertainty.

Perception

Perception: From Raw Data to Meaning

Perception transforms sensor signals into useful world models.

Typical outputs:

- **Detection:** What objects are present?
- **Localization:** Where are they?
- **Classification:** What category does each object belong to?
- **Tracking:** How are objects moving over time?

Challenge: Real-world data is noisy, incomplete, and ambiguous.

Levels of Visual Perception

- **Low-level vision:** edges, corners, gradients, keypoints.
- **Mid-level vision:** regions, contours, depth cues, motion cues.
- **High-level vision:** object categories, scene semantics, intent.

Classical examples:

- Stereo vision: estimate depth from disparity.
- Optical flow: estimate motion from frame-to-frame changes.

Modern systems combine classical geometry with learned models.

Perception Under Uncertainty

Why uncertainty appears:

- Sensor noise and calibration errors.
- Occlusion (objects partially hidden).
- Changing illumination and weather.
- Dynamic scenes with many moving agents.

Common strategies:

- Probabilistic filtering (e.g., Kalman-like updates).
- Multi-sensor fusion for robustness.
- Temporal smoothing and confidence estimates.

Sensors

Sensors provide observations used for state estimation and planning.

Key categories:

- **Vision:** RGB/mono/stereo cameras.
- **Range:** LiDAR, sonar, infrared depth sensors.
- **Motion:** IMU (accelerometer + gyroscope), wheel encoders.
- **Global position:** GPS/GNSS.
- **Active safety sensing:** radar for distance and velocity.

Sensor Trade-offs

Sensor	Strength	Limitation
Camera	Rich semantic detail, low cost	Sensitive to lighting and weather
LiDAR	Accurate 3D distance measurements	Expensive, lower semantic detail
Radar	Works in fog/rain, good velocity estimation	Lower spatial resolution
IMU	Fast motion updates	Drift over long duration
GPS	Global localization outdoors	Weak/absent indoors or urban canyons

Takeaway: No single sensor is enough; practical systems fuse multiple sensors.

Sensor Fusion (Basic Idea)

Goal: Combine complementary sensor information to improve reliability.

Example in autonomous driving:

- Camera: recognizes traffic signs and lane markings.
- LiDAR: provides geometric obstacle boundaries.
- Radar: estimates relative speed of nearby vehicles.
- GPS + IMU: provides global position and smooth local motion estimate.

Fusion often improves robustness when one modality fails.

Image Classification (Basic Idea)

What Is Image Classification?

Task: Assign an image to one label from a predefined set.

Example classes: {cat, dog, car, pedestrian, stop-sign}.

Pipeline (simplified):

1. Collect labeled images.
2. Preprocess (resize, normalize, augment).
3. Train model to map pixels $\rightarrow$ class probabilities.
4. Predict class with highest probability.

From Features to Deep Learning

Classical approach:

- Hand-crafted features (edges, texture descriptors).
- Classifier (e.g., SVM, logistic regression).

Modern approach: CNNs

- Learn hierarchical features directly from data.
- Early layers: edges and color blobs.
- Deeper layers: parts and object-level concepts.

Why CNNs work well: local receptive fields + shared weights + translation tolerance.

Classification Output and Decision

For image x , model outputs class probabilities:

$$p(y = k \mid x), \quad k = 1, \dots, K$$

Predicted label:

$$\hat{y} = \arg \max_k p(y = k \mid x)$$

Confidence matters:

- High confidence: model is certain.
- Low confidence: defer to human or safer fallback policy.

In safety-critical systems, decision policies include uncertainty thresholds.

Beyond Classification

In robotics, classification alone is not enough.

Additional vision tasks:

- **Object detection:** classify + localize with bounding boxes.
- **Semantic segmentation:** classify each pixel.
- **Depth estimation:** infer distance from images.
- **Pose estimation:** infer orientation and key points.

These tasks support navigation, manipulation, and human-robot interaction.

AI in Autonomous Systems

AI Stack in Autonomous Systems

A practical autonomy stack often includes:

- **Perception models:** detect lanes, vehicles, pedestrians.
- **Prediction models:** forecast other agents' motion.
- **Planning models:** choose safe and efficient trajectories.
- **Control algorithms:** track trajectory under dynamics constraints.
- **Monitoring/safety layer:** override unsafe actions.

System-level reliability matters more than any single model's accuracy.

Example: Autonomous Driving Loop

Step	Typical Function
Perceive	Detect lanes, signs, drivable area, and dynamic obstacles.
Localize	Estimate ego pose relative to map and road geometry.
Predict	Estimate trajectories of nearby vehicles and pedestrians.
Plan	Select route and short-horizon safe maneuver.
Control	Convert to steering, throttle, and braking commands.
Verify	Apply safety checks and fail-safe fallback if needed.

Case Insight from Core AI Texts

Introductory AI texts discuss robotic vehicles as complete AI agents:

- Driverless platforms combine **cameras, radar, and LiDAR**.
- Perception and planning operate continuously in closed loop.
- Sensor design strongly influences what can be perceived and controlled.

This supports a key principle:

Better autonomy comes from strong integration of sensors, perception, planning, and control.

Challenges in Real Deployment

Technical challenges:

- Long-tail edge cases are hard to cover in training data.
- Domain shift: weather, geography, and sensor aging.
- Real-time compute constraints on embedded hardware.

Safety and ethics challenges:

- Reliable uncertainty estimation and fallback behavior.
- Bias and fairness in perception models.
- Accountability and regulation in autonomous decision making.

Evaluation Metrics (Overview)

Robotics and vision systems are evaluated at multiple levels:

Perception:

- Classification accuracy, precision, recall, F1.
- Detection mAP (mean Average Precision).

Autonomy/system:

- Collision rate and safety intervention count.
- Lane-keeping and trajectory tracking error.
- Mission success and average completion time.

The final objective is safe, robust behavior in real environments.

Summary and Next Steps

What we covered:

- Robotics as a sense-think-act AI system.
- Perception fundamentals and uncertainty.
- Sensor types and the need for fusion.
- Basic image classification and CNN intuition.
- Role of AI in autonomous systems and deployment risks.

Key takeaway: Robust autonomy requires both strong models and strong systems engineering.

Use the next slides for quick self-assessment and discussion.

Self-Assessment Questions I

1. Describe the *sense-think-act loop* and explain why all three components are essential in robotics.
2. Why is sensor fusion necessary? Give an example of how combining camera and LiDAR improves robustness.
3. A CNN trained on sunny outdoor images shows poor performance on rainy images. What problem is this, and how might you address it?

Self-Assessment Questions II

4. In autonomous driving, what trade-offs exist between classification confidence thresholds and system safety?
5. Name at least two real-world challenges (not in training) that make robust autonomous systems hard to deploy.